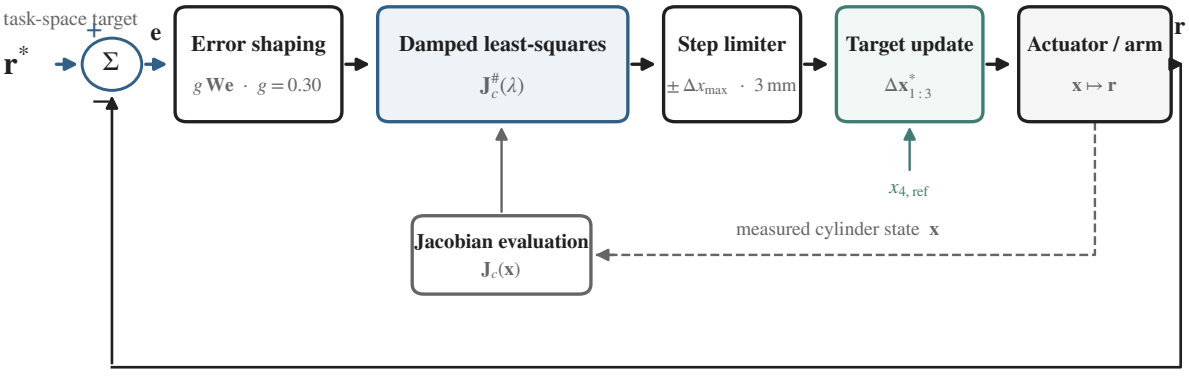
task-space target
r*
+
Σ
e
Error shaping
g We · g = 0.30
Damped least-squares
J#c(λ)
Step limiter
± Δxmax · 3 mm
Target update
Δx*1 : 3
Actuator / arm
x ↦ r
r
x4, ref
Jacobian evaluation
Jc(x)
measured cylinder state x
measured cutter position r